

Assignment #1 — due Wednesday, Sep. 24, 2025, at 11:59PM

- *Submit your homework on Canvas.
- *No late homework will be accepted for credit.
- *Append the codes you used to your submission.
- *Using AI tool is allowed for coding problems, but needs to be declared in your submission.

Problem 1: Basic Knowledge (100 points)

1. “Overfitting occurs only when training error is zero. ” Is the claim *True or Wrong*? Explain!
2. When K increases, does the variance of KNN decrease or increase? Explain!
3. Consider classification problem with the response variable Y taking possible values from $\{1, \dots, K\}$ and denote $q_j(x) = \mathbb{P}(Y = j|X = x), \forall j = 1, \dots, K$ the probability that $Y = j$ given $X = x$. What is the expected misclassification error of the *Bayes Classifier*? You can use $p(x)$ to represent the probability density function of X .
4. Show that the sum of residuals in simple linear regression is zero. How about in multiple linear regression?
5. Given two linear models with the same number of parameters, can we use R square to compare the performance of the two models, or must we use the adjusted R square? Explain!.
6. Explain the advantage and disadvantage of linear regression and KNN regression.

Problem 2: Investigation of Life Expectancy (100 points)

Dataset: Life_Expectancy_Data.csv

Description: The Global Health Observatory (GHO) data repository under World Health Organization (WHO) keeps track of the health status as well as many other related factors for all countries. The data-sets are made available to public for the purpose of health data analysis. The dataset related to life expectancy, health factors for 193 countries has been collected from the same WHO data repository website and its corresponding economic data was collected from United Nation website. Among all categories of health-related factors only those critical factors were chosen which are more representative. The final dataset consists of 22 Columns and 2938 rows which meant 20 predicting variables. The meaning of these predictors are as follows.

Year: 2000-2015

Status: Developed or Developing

Life expectancy: Life Expectancy in age

Adult Mortality: Adult Mortality Rates of both sexes (probability of dying between 15 and 60 years per 1000 population)

infant deaths: Number of Infant Deaths per 1000 population

Alcohol: Alcohol, recorded per capita (15+) consumption (in litres of pure alcohol)

percentage expenditure: Expenditure on health as a percentage of Gross Domestic Product per capita(%)

Hepatitis B: Hepatitis B (HepB) immunization coverage among 1-year-olds (%)

Measles: Measles - number of reported cases per 1000 population

BMI: Average Body Mass Index of entire population

under-five deaths: Number of under-five deaths per 1000 population

Polio: Polio (Pol3) immunization coverage among 1-year-olds (%)

Total expenditure: General government expenditure on health as a percentage of total government expenditure (%)

Diphtheria: Diphtheria tetanus toxoid and pertussis (DTP3) immunization coverage among 1-year-olds (%)

HIV/AIDS: Deaths per 1 000 live births HIV/AIDS (0-4 years)

GDP: Gross Domestic Product per capita (in USD)

Population: Population of the country

thinness 1-19 years: Prevalence of thinness among children and adolescents for Age 10 to 19 (%)

thinness 5-9 years: Prevalence of thinness among children for Age 5 to 9(%)

Income composition of resources: Human Development Index in terms of income composition of resources (index ranging from 0 to 1)

Schooling: Number of years of Schooling(years)

Goal: Fit linear regression models to answer the following questions. Note that there are 193 countries which will be treated as factors. Although it is an important factor, please ignore the predictor “Country”. Otherwise, there will be around 200 predictors.

1. Report the summary of the linear model. What are the predicting variables actually affecting the life expectancy? Justify your answer based on the outputs of linear regression model.
2. Construct the 95% confidence intervals for the coefficient of “Adult Mortality” and “HIV/AIDS”. Are you confident that these predictors have negative impact on the life expectancy? Explain why.
3. Construct the 97% confidence intervals for the coefficient of “Schooling” and “Alcohol”. Explain how these predictors impact the life expectancy.
4. Based on the p-values, which are the top-seven most influential predictors? Use these predictors to fit a smaller model and report the summary.
5. Use the smaller model to predict the life expectancy if a new observation is given with *Year=2008, Status=Developed, Adult Mortality=125, infant deaths=94, Alcohol=4.1, percentage expenditure=100, Hepatitis B=20, Measles=13, BMI=55, under-five deaths=2, Polio=12, Total expenditure=5.9, Diphtheria=12, HIV/AIDS=0.5, GDP=5892, Population=1.34 × 10⁶, Income composition of resources=0.9, Schooling=18*. Report the 99% confidence interval for your prediction.
6. Use AIC to compare the full model and the smaller model.

Problem 3: Implementing KNN regression (100 points)

Dataset: boston_housing_train.csv and boston_housing_test.csv

Goal: use any programming language to implement the KNN regression to predict *medv*. You cannot use default KNN regression function in R or Python. Document your codes carefully!

Submit the *source code with a detailed instructions on how to use your code*. Then report

1. The MSE and the corresponding running time on test data for different values of *K*. Report the best value of *K*.

2. Standardize the predictor variables (excluding the response variable) in both training and test data. Report The MSE and the corresponding running time on test data for different values of K . Report the best value of K .
3. Does standardization improve the performance?